Li 2019 Melanoma — analysis summary

RUN 3 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony θ-sweep read on the regenerated embedding + per-neighbourhood iLISI (never a scalar): the kNN graph is ONE connected component at every θ∈{2,5,10,20} × n_pcs∈{20,30,40}; iLISI rises 1.30→1.80/3 and plateaus (θ=10 settled). Each of the 3 patients spans BOTH density lobes (p21 43/57, p25 59/41, p26 58/42) — a patient-shared axis, not an archipelago. The two visual poles are biologically identical (signatures + cell cycle equal), so they are density poles of one manifold.

Gate 2 — module geography → GO (non-branching)

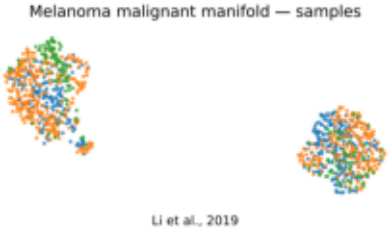
Low-signal melanoma (raw means Gast −0.015, pSom +0.016, Crest −0.002). Gast & pSom co-localise at the origin and decline along pseudotime (Gast ρ=−0.47; pSom flat ρ=+0.05) opposed to a terminal neural-crest end that rises (Crest ρ=+0.59; Crest-high decile pt=1.00). Crest anti-correlated with both in space (Crest↔Gast −0.37, Crest↔pSom −0.43); joint Gast&pSom-high enrichment 0.20 (<1) — an inverse terminal gradient, not a co-expression hub. Crest peaks at the terminus, not on a Gast arm — no inversion flag.

Gate 3 — hub cross-repression → NA

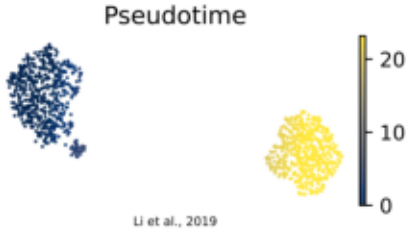
Single monotone axis — no bifurcation (curve: 2 tips, 0 forks, fold-free self-approach 2.99; a PPT tree on the same manifold also yields 0 forks) and no Gast/pSom co-expression hub, so there is no branch pair to test for Gast↔pSom cross-repression.

STEP FIGURES

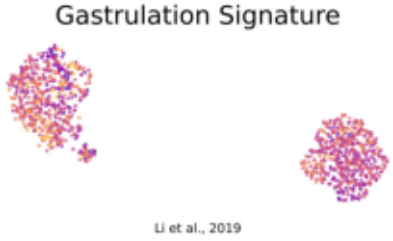
Manifold — samples



Pseudotime



Gastrulation



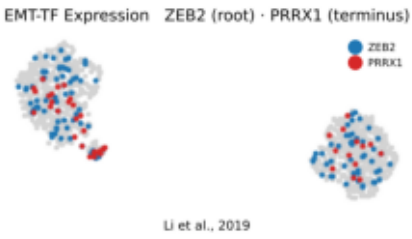
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Melanoma malignant cells (Li et al., 2019), MARS-seq — an immune-focused atlas: 1,160 of 78,336 cells are malignant; ≥56-cell dominant-gap floor keeps 3/20 samples → 1,082 cells on the trajectory.

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous, sample-integrated gastrulation → terminal-neural-crest axis rooted at the Gast/pSom-high origin. Gate 1 GO / Gate 2 GO / Gate 3 NA; no Puram-style hub-and-spoke bifurcation (curve 0 forks), so Gate 3 has no hub to test. EMT-TFs ZEB2 (body) / PRRX1 (sparse), detection-gated on this shallow dataset.

Parameter choices

Sample threshold	56 cells (dominant multiplicative gap 154→21, ratio 7.3×); 3/20 samples, 1,082 cells	**Overdispersed genes**	217 (scFates find_overdispersed); no sc.pp.scale before PCA
Integration	Harmony θ=10 (iLISI plateau; strongly patient-private melanoma programs)	**Embedding**	n_pcs=30 (mid robust 20–40), n_neighbors=15; single elongated axis (aspect 7.5)
Geometry	Non-branching scFates curve, N=20 λ=0.3 μ=100; 2 tips/0 forks, fold-free (SA 2.99); PPT 0 forks	**Root**	tip 0 = Gast/pSom-high, Crest-low origin
Module means	Gast −0.015 / pSom +0.016 / Crest −0.002 (low-signal melanoma)	**EMT-TFs**	ZEB2 (body) ↔ PRRX1 (sparse terminus); VIM excluded